

LESSON 6-7 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Factors and Greatest Common Factor**

Maryland Standard 6.NS.B.4 • Learning target – I can list the factors of two numbers and find their greatest common factor.

REMEMBER

- A **factor** divides a number with no remainder. Find factors in pairs: $18 = 1 \times 18 = 2 \times 9 = 3 \times 6$.
- The **greatest common factor** (GCF) is the greatest factor that two numbers share.
- Use the GCF to make the **greatest number of identical groups** with nothing left over.

WORKED EXAMPLE

Find the GCF of 24 and 36.

Step 1 Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24

Step 2 Factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36

Step 3 Common factors: 1, 2, 3, 4, 6, 12. The greatest is 12.

Answer GCF = 12

WORD BANK

factor – a number that divides another number evenly

common factor – a factor both numbers have

GCF – the biggest factor both numbers share

prime – a number with only two factors: 1 and itself

PRACTICE

show your work • label units

1. List all the factors of each number.

Step 1 Start with 1 and the number itself.

Step 2 Try 2, 3, 4, ... and write each pair that multiplies to the number.

a. 30: _____

b. 28: _____

c. 45: _____

d. 17: _____

e. 48: _____

f. 42: _____

2. Find the GCF of each pair.

Step 1 List the factors of each number.

Step 2 Circle the factors that are in both lists.

Step 3 Pick the biggest circled factor.

a. 12 and 20: ____

b. 18 and 27: ____

c. 16 and 40: ____

d. 14 and 25: ____

e. 36 and 48: ____

f. 30 and 45: ____

- 3.** Write the prime factorization of each number. Use exponents.

a. $36 = \underline{\hspace{2cm}}$

b. $50 = \underline{\hspace{2cm}}$

c. $72 = \underline{\hspace{2cm}}$

- 4. Snack Bags.** Mr. Diaz has 36 granola bars and 54 juice boxes. Every bag must be the **same**. Nothing is left over. What is the **greatest** number of bags? What goes in each bag?

Step 1 Find the GCF of 36 and 54. That is the number of bags.

Step 2 Divide each number by the GCF to fill one bag.

Show your work.

Answer _____

- 5. Find the Error.** Hana says the GCF of 15 and 28 is 0 because they share no factors. Explain her mistake and give the GCF.

Frame Hana is wrong because every number has the factor _____. So the GCF of 15 and 28 is _____.

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help

☐ I need a little help

☐ I have a question about #_____

AT HOME Ask someone at home to name two numbers. Find their GCF together.

LESSON 6-7 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Multiples and Least Common Multiple**

Maryland Standard 6.NS.B.4 • Learning target – I can list multiples of two numbers and find their least common multiple.

REMEMBER

- **Multiples** of a number come from skip-counting: 6, 12, 18, 24, ...
- The **least common multiple** (LCM) is the least number that is a multiple of both.
- Use the LCM when events **happen together again**, or to buy packs for **equal amounts**.

WORKED EXAMPLE

Find the LCM of 8 and 12.

Step 1 Multiples of 12: 12, 24, 36, ...**Step 2** 12 is not a multiple of 8. 24 is ($8 \times 3 = 24$).**Answer LCM = 24****WORD BANK****multiple** – a number you get when you skip-count**common multiple** – a multiple both numbers have**LCM** – the smallest multiple both numbers share**PRACTICE***show your work • label units*

- 1.**
- Find the LCM of each pair.

Step 1 Skip-count by the bigger number.**Step 2** Stop at the first number the smaller number also divides.**a.** 3 and 7: ____**b.** 4 and 10: ____**c.** 6 and 8: ____**d.** 5 and 15: ____**e.** 9 and 12: ____**f.** 10 and 25: ____

- 2. Cookout.**
- Hot dogs come in packs of 8. Buns come in packs of 10. Malik wants the
- same number**
- of hot dogs and buns. What is the
- fewest**
- packs of each he can buy?

Step 1 Find the LCM of 8 and 10.**Step 2** Divide the LCM by 8 and by 10 to count packs.*Show your work.***Answer** _____

- 3. Blinking Lights.**
- One light blinks every 6 seconds. Another blinks every 9 seconds. They just blinked together. In how many seconds will they blink together again?

Step 1 List multiples of 6 and of 9.**Step 2** Find the first number on both lists.**Answer** _____ seconds

4. Two runners start together. One laps the track every 4 minutes, the other every 6 minutes. Which tells when they are next at the start together?

☐ GCF of 4 and 6

☐ LCM of 4 and 6

5. **Reason.** Is the LCM of two numbers always their product? Use 4 and 5, and 4 and 6, to explain.

Frame LCM of 4 and 5 = ____; $4 \times 5 =$ ____. LCM of 4 and 6 = ____; $4 \times 6 =$ ____. So the LCM is ____ the product.

TEST PRACTICE

choose the best answer

6. What is the LCM of 6 and 15?

A 3

B 30

C 90

D 21

7. Which pair of numbers has an LCM of 24?

A 4 and 6

B 6 and 8

C 2 and 12

D 24 and 48

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____